Frames, Relativity and Time Dilation

Worked answer key: question, diagram, explanation with reasons, final answer

Conventions used throughout

• **Frames:** an **inertial** frame moves at constant velocity (not speeding up, slowing down or turning). A **non-inertial** frame is accelerating, for example a car that is turning.

• **Time notation:** Δt_s is the time measured by an observer for whom the clock is at rest (stationary relative to the clock); this is the proper time. Δt_m is the time measured by an observer for whom the clock is moving.

• **Constants:** $c = 3.00 \times 10^8$ m/s is the speed of light. v is the speed of the clock (or object) relative to the observer. γ (the Greek letter gamma) is short for $1/\sqrt{(1 - v^2/c^2)}$, so $\Delta t_m = \gamma \Delta t_s$.

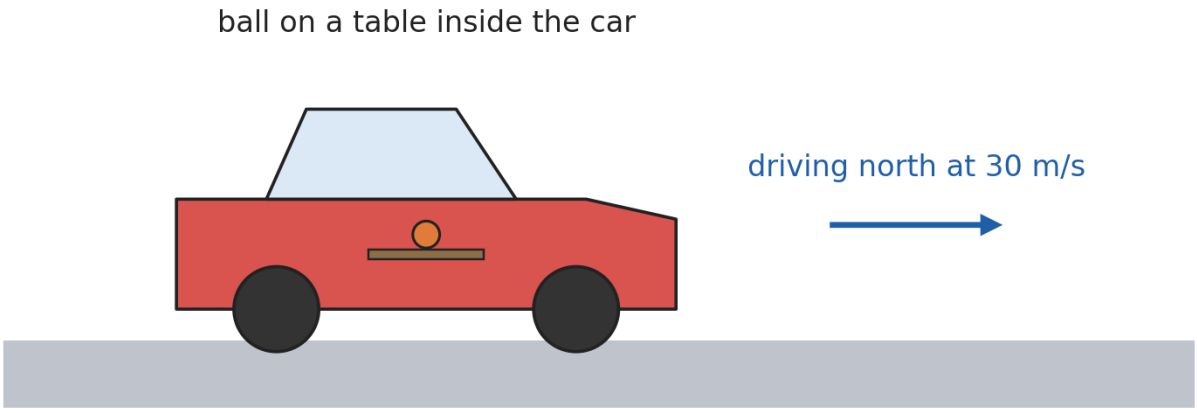
FRAME EXAMPLE

Car turns: ball in two frames

A ball is sitting on a level table in a car that's moving directly north with a constant speed of 30 m/s. The ball remains at rest with respect to the car and remains moving at 30 m/s [N] with respect to the road.

In a non-inertial frame of reference what happens to the ball if the car suddenly turns to the right? Draw the direction the ball moves with respect to the car.

Picture the situation



A ball sits on a smooth table in a car driving north at 30 m/s. Then the driver turns right.

Step 1 – Before the turn

While the car drives straight at a steady speed, the car is an inertial frame (it isn't accelerating), and so is the road. Someone in the car sees the ball sitting still. Someone standing by the road sees the ball moving north at 30 m/s along with the car. Both are correct: they are just describing it from different frames.

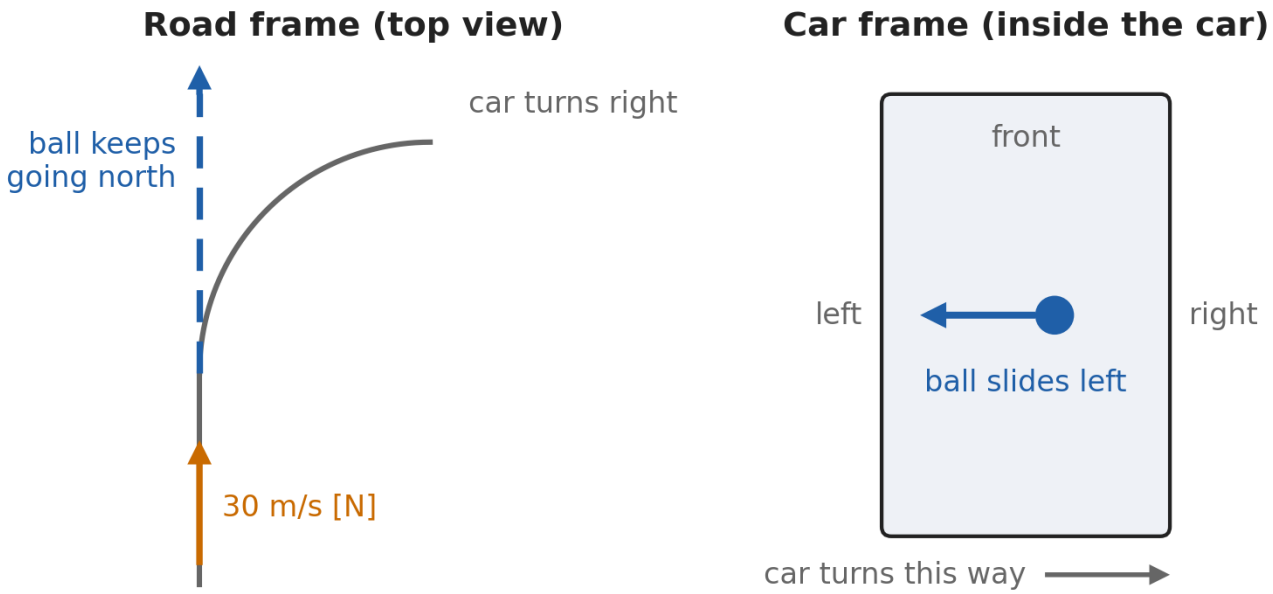
$$\vec{v}_{ball/car} = 0 \qquad \vec{v}_{ball/road} = 30 \text{ m/s [N]}$$

Step 2 – During the turn

When the car turns, think about it from the road first, because the road is still an inertial frame and Newton's laws work normally there. Nothing pushes the ball sideways (the table is level and we ignore friction), so by Newton's first law the ball just keeps going straight north at 30 m/s. The car is what changes: it swings to the right (toward the east), out from under the ball.

Now look from inside the car. The car is turning, so it is accelerating and is a non-inertial frame. From there, the ball seems to slide toward the car's **left** side (the outside of the turn), even though nothing is pushing it. That is the same feeling you get of being thrown sideways when a car turns.

Diagram: the same motion in two frames



The ball moves toward the left side of the car (the outside of the turn). Nothing pushes it left: the car turns right underneath it.

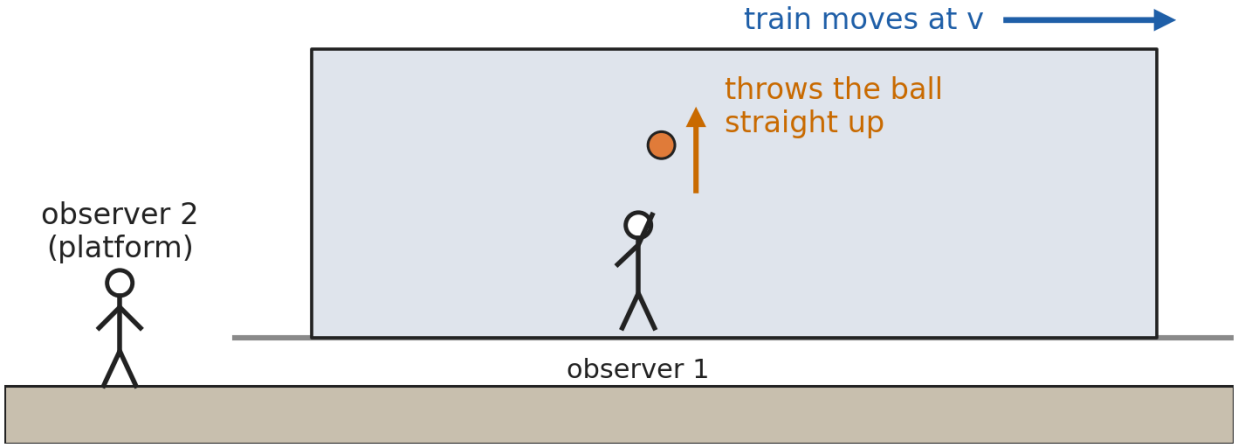
EXAMPLE

Ball seen from train and platform

(a) When observer 1 throws a ball upward, he observes that the ball's motion is purely along the vertical (y) direction.

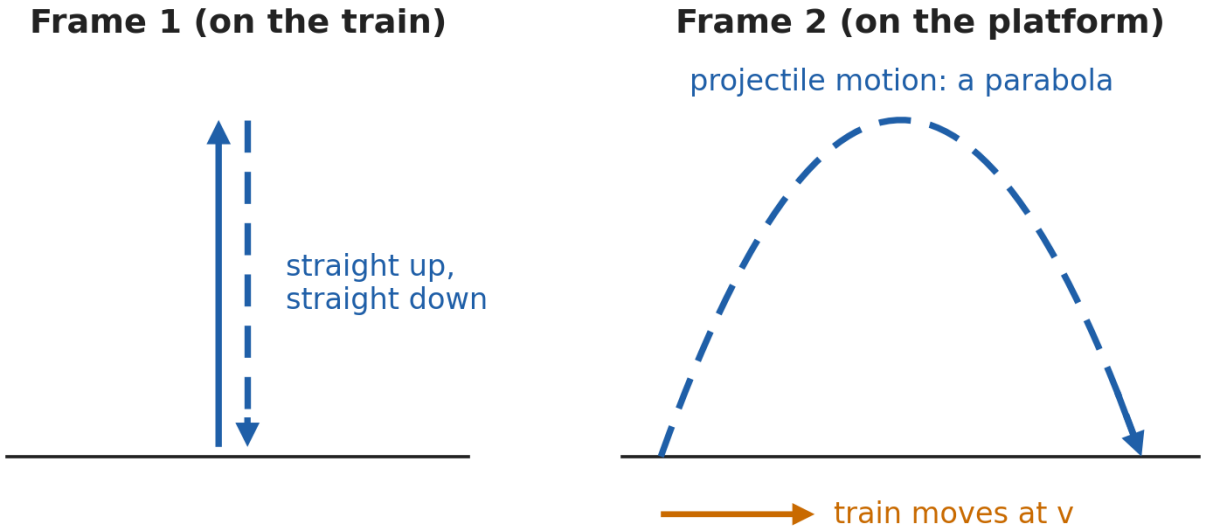
(b) When the railway car for observer 1 has a speed v relative to observer 2, the ball appears to undergo projectile motion with a displacement along both x and y in reference frame 2. However, as viewed by observer 1 using his reference frame and coordinates x and y, the ball's motion is still purely vertical, as in (a).

Picture the situation



Observer 1 rides in a railway car and throws a ball straight up. Observer 2 stands on the platform as the car goes by.

Diagram: one throw, two frames



Why observer 1 sees straight up and down

Observer 1 moves along with the ball before it is thrown, so relative to him the ball starts with no sideways motion at all. The railway car moves at constant velocity (an inertial frame), so nothing about the steady motion changes what he sees. Gravity only changes the ball's up-and-down speed, so he sees it go straight up and fall straight back into his hand.

$$\text{train frame: } v_x = 0, \quad a_y = -g$$

Why observer 2 sees a parabola

From the platform, the ball was already moving forward at the train's speed v before it was thrown (it was in observer 1's hand on the moving train). So it leaves with an upward velocity *and* a forward velocity v. The forward part stays the same while gravity changes the vertical part. That is exactly projectile motion, so observer 2 sees a parabola.

$$\text{platform frame: } v_x = v \text{ (constant)}, \quad a_y = -g$$

Both observers are right. The path depends on the frame, but the physics (the same g, the same laws) is the same in both inertial frames.

PRINCIPLES

Two principles of special relativity

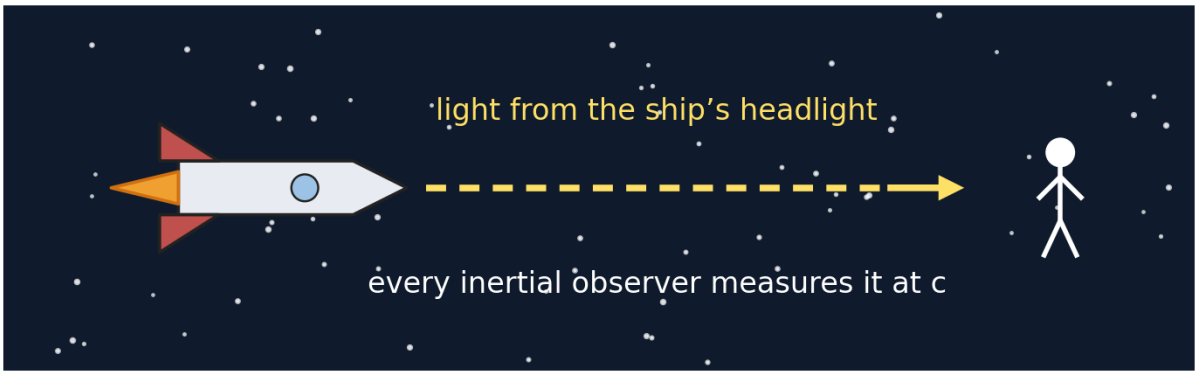
The Principle of Relativity

- The laws of physics are the same in all inertial frames of reference. No physics experiment can ever determine whether you are at rest or moving at a constant velocity.

The Speed of Light Principle

- There is at least one inertial frame of reference in which, for an observer at rest in this frame of reference, the speed light, c, in a vacuum is independent of the motion of the source of the light.

Picture the situation



However fast the ship moves, anyone moving at constant velocity who measures its light gets the same speed, c.

What the principle of relativity means

Imagine a train moving perfectly smoothly at a constant speed, with the blinds down. A dropped ball falls straight down, a pendulum swings normally, a light clock ticks normally, exactly as if the train were parked. No experiment inside can tell you whether you are moving. You can only notice motion when the velocity changes (speeding up, braking or turning).

What the speed-of-light principle means

Combined with the first principle, it means every inertial observer measures the same speed for light in a vacuum, $c = 3.00 \times 10^8$ m/s, whether the source or the observer is moving. This is very different from everyday motion: a ball thrown forward from a moving train is faster for someone on the ground, but light from a moving ship is *not* faster. Ordinary velocity addition (ship's speed + light's speed) does not work for light. Because everyone must get the same c, something else has to give: measured times and lengths. That is where time dilation and length contraction come from.

Physics laws are the same in every inertial frame, and every inertial observer measures light in a vacuum at c, even when the source moves.

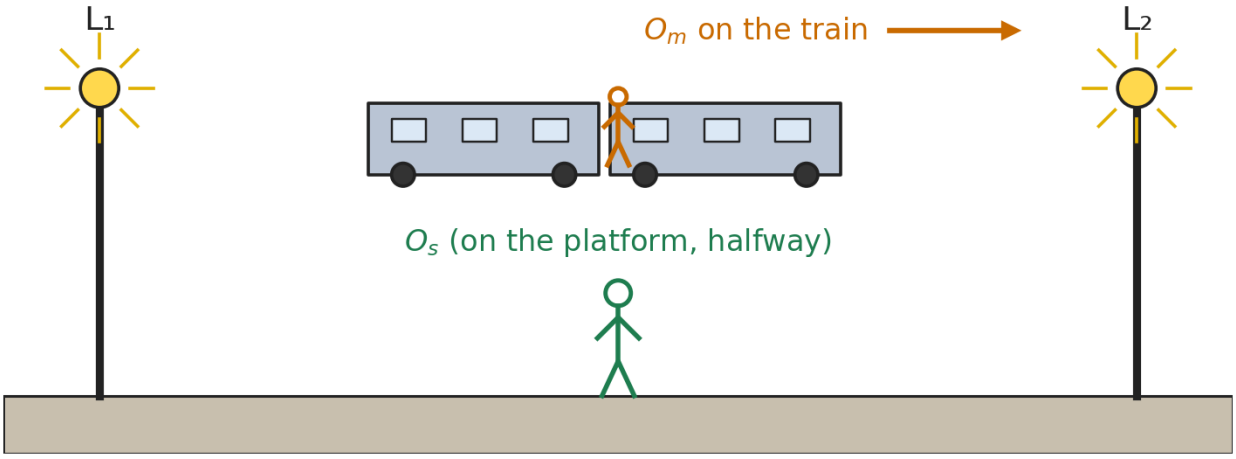
SIMULTANEITY

Two lamps and a moving train

Two events that are simultaneous in one reference frame are not necessarily simultaneous in another reference frame that's moving relative to the first.

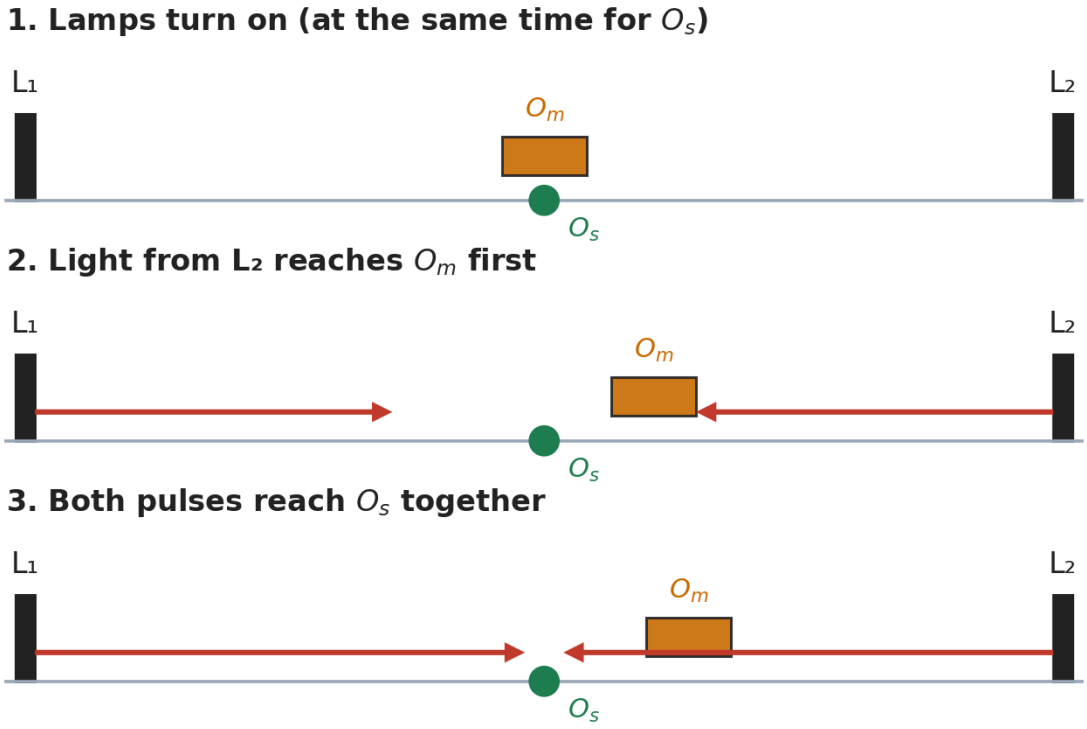
An observer O_s, stationary in the inertial frame of Earth, is standing on a railway platform at the midway point between two lampposts, L_1 and L_2. The lampposts are connected to the same circuit, ensuring that, at least from the viewpoint of an inertial observer, they turn on at the same time.

Picture the situation



Two lamps switch on together. O_s stands still on the platform halfway between them. A second observer, O_m, rides a train toward L_2 and is right beside O_s when the lamps turn on.

Diagram: three moments, seen from the platform



The train observer O_m moves toward L_2, so he meets L_2's light before L_1's.

Step 1 – The platform observer

O_s stands still exactly halfway between the lamps. Light from each lamp travels the same distance to her at the same speed, c. So if both flashes reach her at the same moment, they must have started at the same moment. She concludes the lamps turned on at the same time.

Step 2 – The train observer

O_m is moving toward L_2, so he runs into L_2's light before L_1's light catches up with him (see the diagram). Everyone agrees on this order, because both lights arrive at the same place (where O_m is). Now reason from his point of view. In his own frame *he* is at rest. The two flashes started from points equally far in front of and behind him, and both pulses travel at c (light has the same speed for everyone). If the lamps had flashed together, the light would have reached him together. It didn't: L_2's came first. So he concludes L_2 turned on **before** L_1.

Both observers reason correctly. They disagree because "at the same time" depends on the frame of reference.

For O_s the lamps turn on at the same time; for O_m L_2 turns on first. Simultaneity depends on the frame of reference.

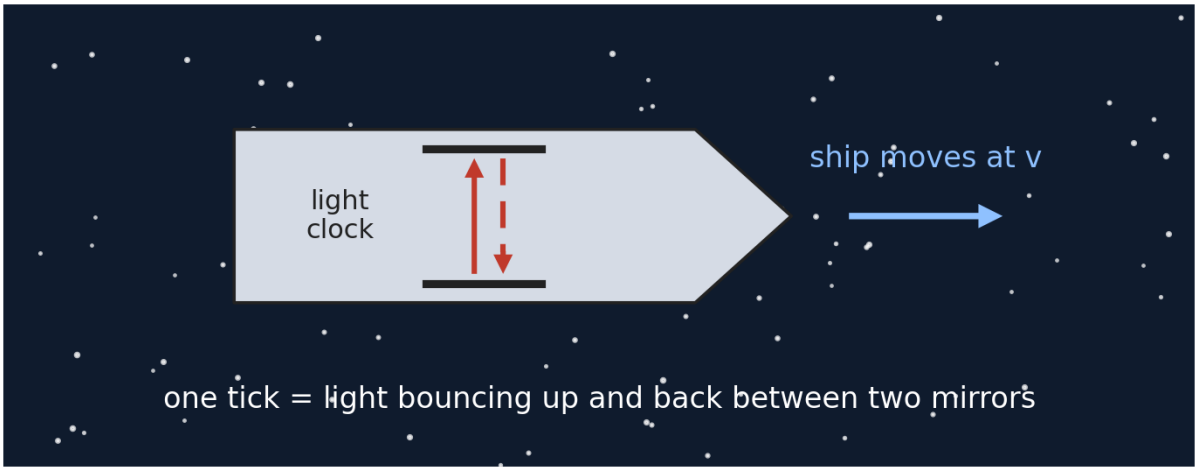
LIGHT CLOCK

Derive time dilation with a light clock

We can determine the relationship between Δt_s and Δt_m using the Pythagorean theorem.

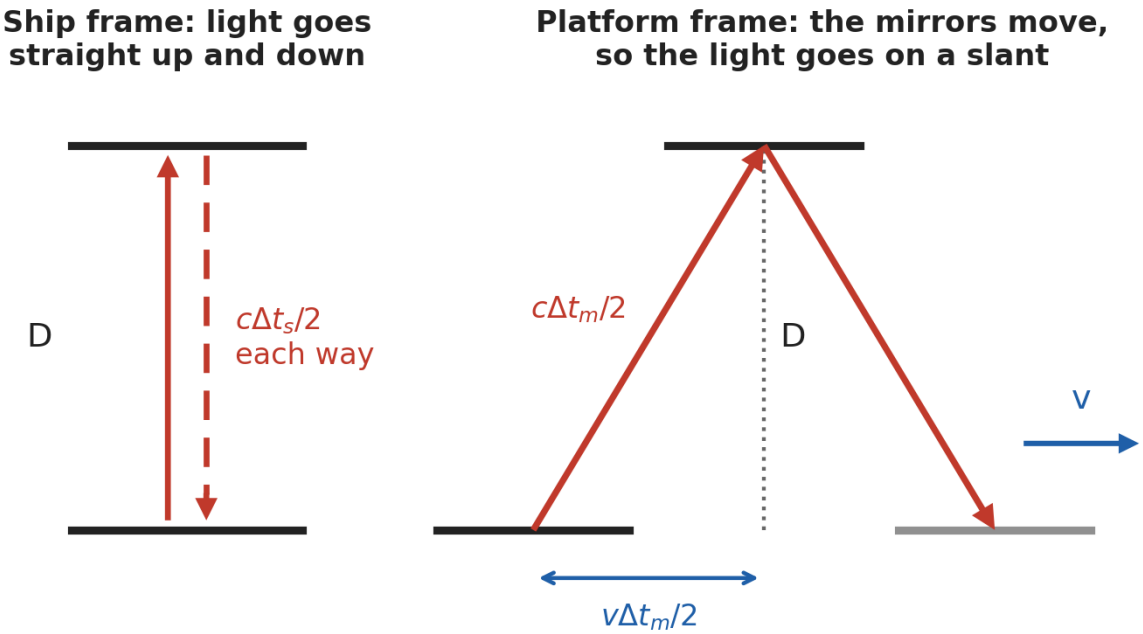
Isolate for Δt_m to see how it compares to Δt_s.

Picture the situation



A light clock on a spaceship: each tick is a light pulse going up to a mirror and back down. The ship flies past a platform at speed v.

Diagram: one tick of the light clock in each frame



Step 1 – Ship frame

Start with someone on the ship. For them the clock isn't moving, so the light goes straight up a distance D and straight back down. Call the time for one round trip Δt_s (s for "stationary" relative to the clock). Half of that, $\Delta t_s/2$, is the time to go up once.

Formula: distance = speed × time

Light goes up a distance D at speed c in half a tick: $D = c\frac{\Delta t_s}{2}$

Step 2 – Platform frame

Now watch the same tick from the platform. Call the time for one round trip Δt_m (m for "moving" clock). v is the ship's speed relative to the platform. While the light goes up, the whole ship moves sideways, so the light has to travel on a slant to reach the top mirror. In half a tick: the ship moves sideways $v\Delta t_m/2$, and the light travels $c\Delta t_m/2$ along the slant (it still moves at c, by the second principle). The height D is the same as before, because it is perpendicular to the motion. These three lengths make a right triangle (see the diagram), so Pythagoras applies. Then replace D using Step 1:

Formula (Pythagoras): $(\text{hypotenuse})^2 = (\text{side 1})^2 + (\text{side 2})^2$

$$\left(\frac{c\Delta t_m}{2}\right)^2 = \left(\frac{v\Delta t_m}{2}\right)^2 + D^2$$

Replace D with $\frac{c\Delta t_s}{2}$ from Step 1:

$$\left(\frac{c\Delta t_m}{2}\right)^2 = \left(\frac{v\Delta t_m}{2}\right)^2 + \left(\frac{c\Delta t_s}{2}\right)^2$$

Step 3 – Isolate Δt_m

Now it is just algebra, one move at a time. The goal is to get Δt_m by itself. We take the positive square root at the end because times are positive. “Factor out” means pulling the $\Delta t_m^{\,2}$ that both terms share out in front of a bracket (multiply the bracket back out to check). In the last line, γ is just the short name for $1/\sqrt{(1 - v^2/c^2)}$.

Square each bracket: $\dfrac{c^2\Delta t_m^2}{4} = \dfrac{v^2\Delta t_m^2}{4} + \dfrac{c^2\Delta t_s^2}{4}$

Multiply every term by 4: $c^2\Delta t_m^2 = v^2\Delta t_m^2 + c^2\Delta t_s^2$

Divide every term by c^2 : $\Delta t_m^2 = \dfrac{v^2}{c^2}\Delta t_m^2 + \Delta t_s^2$

Subtract $\dfrac{v^2}{c^2}\Delta t_m^2$ from both sides: $\Delta t_m^2 - \dfrac{v^2}{c^2}\Delta t_m^2 = \Delta t_s^2$

Factor out Δt_m^2 : $\Delta t_m^2\left(1 - \dfrac{v^2}{c^2}\right) = \Delta t_s^2$

Divide both sides by the bracket: $\Delta t_m^2 = \dfrac{\Delta t_s^2}{1 - \dfrac{v^2}{c^2}}$

Square root both sides: $\Delta t_m = \dfrac{\Delta t_s}{\sqrt{1 - \dfrac{v^2}{c^2}}} = \gamma\,\Delta t_s$

$\Delta t_m = \Delta t_s / \sqrt{(1 - v^2/c^2)}$. When $v > 0$ the square root is less than 1, and dividing by a number less than 1 gives a bigger answer, so $\Delta t_m > \Delta t_s$. One tick takes longer by the platform’s clocks, so the moving clock is measured to run slow.

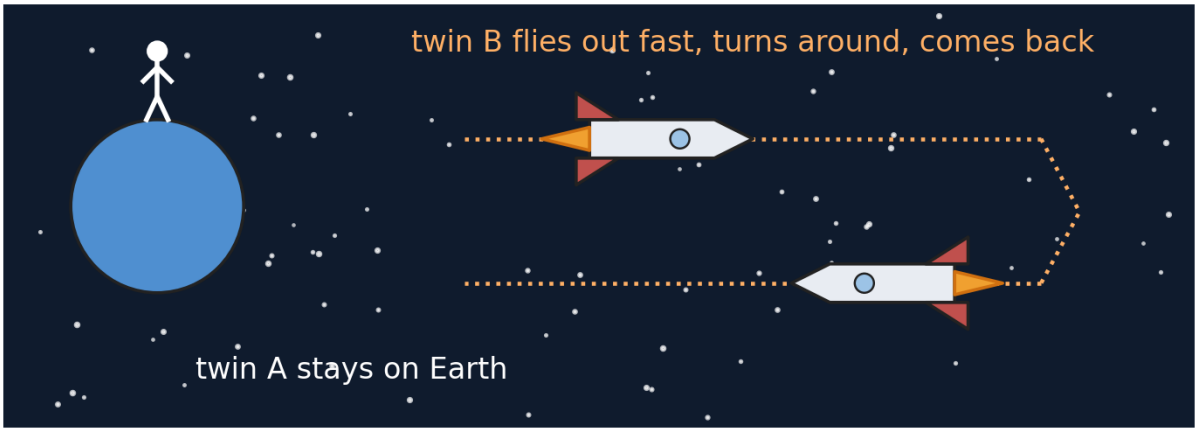
TWIN PARADOX

Which twin is younger?

Explain why people who are beginners at relativity think there is a paradox.

Explain why the two twins are not equivalent observers and therefore there is no paradox. Who is younger?

Picture the situation

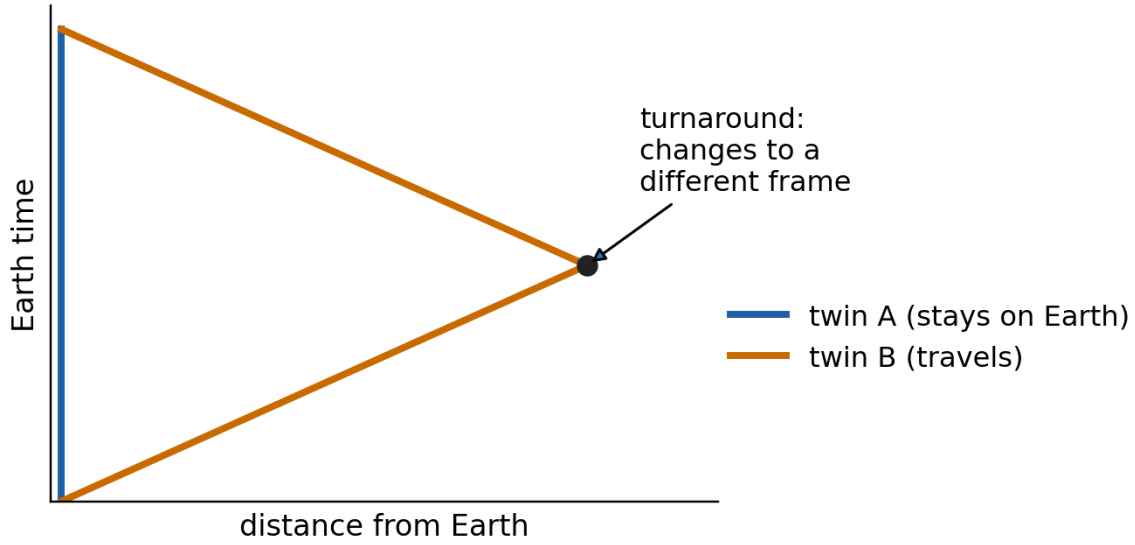


Twin A stays home. Twin B flies out at high speed, turns around, and comes back.

Why it looks like a paradox

From the light clock we know a moving clock runs slow. A beginner says: “From Earth, twin B is moving, so B’s clock runs slow and B is younger. But from the ship, Earth is moving, so A’s clock runs slow and A is younger.” Both twins can’t be younger than each other when they meet again, so it looks like relativity contradicts itself.

Diagram: the two twins’ paths (Earth time vs distance from Earth)



Why there is no paradox

The argument only works if both twins are inertial observers the whole time, and they are not. Twin A stays in one inertial frame (Earth) for the whole trip. Twin B does not: to come home, the ship must slow down, turn around and speed up again. Twin B feels that acceleration (pressed into the seat), and twin A does not. So the two situations are not the same, and the “from the ship’s point of view” argument isn’t valid for the whole trip. Only twin A’s view, a moving clock running slow the whole time, applies to the round trip.

> **Twin B, the travelling twin, is younger at the reunion. The turnaround (acceleration) makes the twins’ situations different, so there is no contradiction.**